

UNIT - 1

What is AI ?

AI is a branch of science that deals with the study and the creation of computer systems that exhibit some form of intelligence.

Artificial Intelligence is a way of **making a computer, a computer-controlled robot, or a software think intelligently**, in the similar manner the intelligent humans think.

AI is accomplished by studying how the human brain thinks and how humans learn, decide, and work while trying to solve a problem, and then using the outcomes of this study as a basis of developing intelligent software and systems.

- Artificial Intelligence (AI) is a branch of Science which deals with helping machines finds solutions to complex problems in a more human-like fashion.
- This generally involves borrowing characteristics from human intelligence, and applying them as algorithms in a computer friendly way.
- A more or less flexible or efficient approach can be taken depending on the requirements established, which influences how artificial the intelligent behavior appears.
- Artificial intelligence can be viewed from a variety of perspectives.
 - ✓ From the perspective of **intelligence** artificial intelligence is making machines "intelligent" -- acting as we would expect people to act.
 - The inability to distinguish computer responses from human responses is called the Turing test.
 - Intelligence requires knowledge
 - Expert problem solving - restricting domain to allow including significant relevant knowledge
 - ✓ From a **business** perspective AI is a set of very powerful tools, and methodologies for using those tools to solve business problems.
 - ✓ From a **programming** perspective, AI includes the study of symbolic programming, problem solving, and search.
 - Typically AI programs focus on symbols rather than numeric processing.
 - Problem solving - achieve goals.
 - Search - seldom access a solution directly. Search may include a variety of techniques.
- Artificial Intelligence is a new electronic machine that stores large amount of information and process it at very high speed.
- The ability to solve problems.
- It is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human intelligence.

Intelligence means:-

- a) Systems that learn new concepts and tasks.
- b) Systems that draw useful conclusions about the world around us.
- c) Systems that can understand a natural language.
- d) Systems that perform other types of feats that require human types of intelligence.

Types of Intelligence

Linguistic intelligence:- The ability to speak, recognize, and use mechanisms of phonology (speech sounds), syntax (grammar), and semantics (meaning).

Musical intelligence:- The ability to create, communicate with, and understand meanings made of sound, understanding of pitch, rhythm.

e.g. Musicians, Singers, Composers

Logical mathematical intelligence:- The ability of use and understand relationships in the absence of action or objects. Understanding complex and abstract ideas.

E.g. Mathematicians, Scientists

Spatial intelligence:- The ability to perceive visual or spatial information, change it, and re-create visual images without reference to the objects, construct 3D images, and to move and rotate them.

e.g. Map readers

Bodily-Kinesthetic intelligence:- The ability to use complete or part of the body to solve problems or fashion products, control over fine and coarse motor skills, and manipulate the objects.

e.g. Players, Dancers

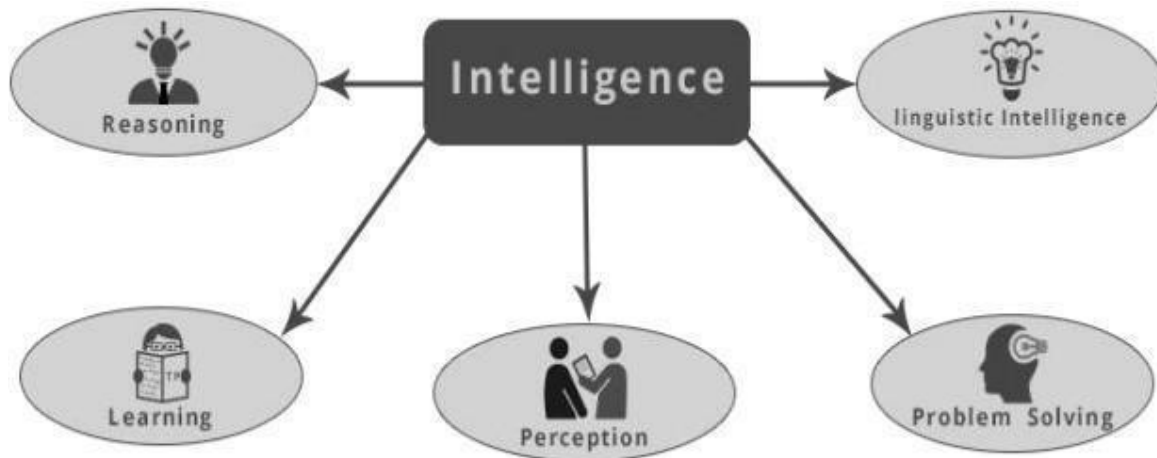
Intra-personal intelligence:- The ability to distinguish among one's own feelings, intentions, and motivations.

e.g. Gautam Buddha

What is Intelligence Composed of?

Intelligence is intangible. It is composed of:

1. Reasoning
2. Learning
3. Problem Solving
4. Perception
5. Linguistic Intelligence



1. **Reasoning:** It is the set of processes that enables us to provide basis for judgment, making decisions, and prediction.
2. **Learning:** It is the activity of gaining knowledge or skill by studying, practicing, being taught, or experiencing something. Learning enhances the awareness of the subjects of the study. The ability of learning is possessed by humans, some animals, and AI-enabled systems.
3. **Problem solving:** It is the process in which one perceives and tries to arrive at a desired solution from a present situation by taking some path, which is blocked by known or unknown hurdles.

Problem solving also includes decision making, which is the process of selecting the best suitable alternative out of multiple alternatives to reach the desired goal are available.

4. **Perception:** It is the process of acquiring, interpreting, selecting, and organizing sensory information.
Perception presumes sensing. In humans, perception is aided by sensory organs. In the domain of AI, perception mechanism puts the data acquired by the sensors together in a meaningful manner.
5. **Linguistic Intelligence:** It is one's ability to use, comprehend, speak, and write the verbal and written language. It is important in interpersonal communication.

Goals of AI

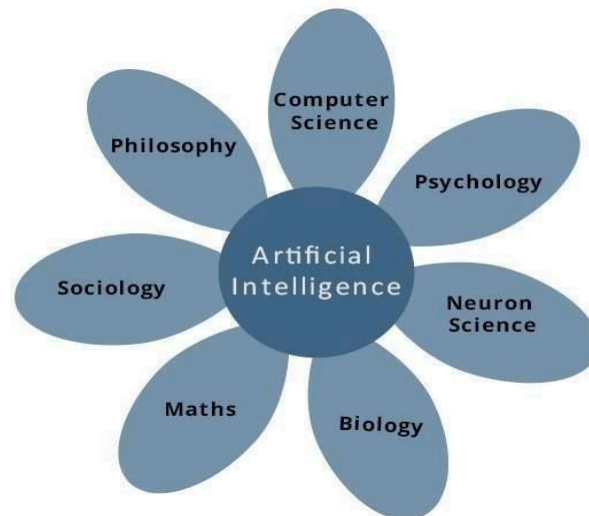
- **To Create Expert Systems** – The systems which exhibit intelligent behavior, learn, demonstrate, explain, and advice its users.
- **To Implement Human Intelligence in Machines** – Creating systems that understand, think, learn, and behave like humans.

What Contributes to AI?

Artificial intelligence is a science and technology based on disciplines such as Computer Science, Biology, Psychology, Mathematics, and Engineering. A major thrust of AI is in the development

of computer functions associated with human intelligence, such as reasoning, learning, and problem solving.

Out of the following areas, one or multiple areas can contribute to build an intelligent system.



- **Examples of AI :-**Following are a few examples of Artificial Intelligence in today's world

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- Voice recognition by a computer system
- Face recognition
- Technology of biometrics
- Driverless vehicles
- Communication with machine, etc.

- **Applications of AI**

- **Gaming:-** AI plays crucial role in strategic games such as chess, poker, tic-tac-toe, etc., where machine can think of large number of possible positions based on heuristic knowledge.
- **Natural Language Processing:-** It is possible to interact with the computer that understands natural language spoken by humans.
- **Expert Systems:-** There are some applications which integrate machine, software, and special information to impart reasoning and advising. They provide explanation and advice to the users.

- **Speech Recognition:-** Some intelligent systems are capable of hearing and comprehending the language in terms of sentences and their meanings while a human talks to it. It can handle different accents, slang words, noise in the background, change in human's noise due to cold, etc.
 - **Handwriting Recognition:-** The handwriting recognition software reads the text written on paper by a pen or on screen by a stylus. It can recognize the shapes of the letters and convert it into editable text.
 - **Intelligent Robots:-** Robots are able to perform the tasks given by a human. They have sensors to detect physical data from the real world such as light, heat, temperature, movement, sound, bump, and pressure. They have efficient processors, multiple sensors and huge memory, to exhibit intelligence. In addition, they are capable of learning from their mistakes and they can adapt to the new environment.
- **Tools used for AI**
 1. Searching
 2. Optimizing
 3. Logic Programming
 4. Automated Reasoning
 5. Probabilistic Methods
 6. Machine learning
 7. Classifiers
 8. Statistical Classifications
 9. Artificial Neural Networks
 10. Intelligent Control
 11. Programming Languages for AI (PROLOG and LISP)
 12. Evaluation in AI

Advantages of AI:

- **Less Errors:** Errors are reduced and the chance of reaching accuracy with a greater degree of precision is a possibility.
- **Faster Decisions:** Using Artificial intelligence, decisions can be taken very fast.
- **Daily Applications:** In today's era, A.I. is used in many applications just like Apple's Siri, Window's Cortana, Google's OK Google. Using these types of applications we can communicate with our device using our voice. Which makes our work easy?
- **No Emotions:** The complete absence of emotions makes machines to think logically and take right decision where in humans emotions are associated with moods that can affect human efficiency.
- **No Breaks:** Unlike humans, machines can work 24*7 without any break. H

- **Medical Applications:** Increasing the integration of A.I. tools in every day medical applications could improve the efficiency of treatments and avoid cost by minimizing the risk of false diagnosis.
- **Taking risks on behalf of humans:** In various situations, Robots can be used instead of Humans to avoid the risks.

Disadvantages of AI:

- **High Costs:** The hardware and software need to get updated with time to meet the latest requirements.
- **Unemployment:** The increasing number of machines leading to unemployment and job security issues. As machines are replacing human resources, the rate of people losing their jobs will increase.
- **Can't think out of box:** Robots can only do the work that they are programmed to do. They cannot act any different outside of whatever algorithm or programming is stored in their internal circuits.
- **Can't feel Compassion and Sympathy:** There is no doubt that machines are much better when it comes to working efficiently but they cannot replace the human connection that makes the team. Machines cannot develop a bond with humans.
- **High dependence on machines:** In today's generation, most of the people are highly dependent on Applications like Siri. With so much assistance from machine, if humans do not need their thinking abilities, these abilities will be gradually decreased. In future with the heavy use of application of artificial intelligence, human may become fully dependent on machines, losing their mental capacities.

AI and its related fields:-

- 1. Chatbots**
- 2. Human Resource Management**
- 3. Artificial Intelligence in e-Commerce**
- 4. AI in Healthcare**
- 5. Artificial Intelligence in Logistics and Supply Chain**
- 6. Streamlined Manufacturing with AI**
- 7. Stock market trading**
- 8. Self-Driving Cars**

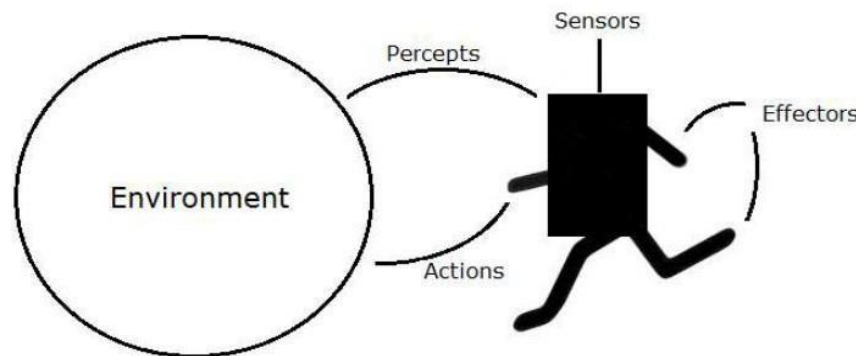
Difference between Human and Machine Intelligence

- Humans perceive by patterns whereas the machines perceive by set of rules and data.
- Humans store and recall information by patterns; machines do it by searching algorithms. For example, the number 40404040 is easy to remember, store and recall as its pattern is simple.
- Humans can figure out the complete object even if some part of it is missing or distorted; whereas the machines cannot correctly.

What are Agent and Environment?

An **agent** is anything that can perceive its environment through **sensors** and acts upon that **environment** through **effectors**.

- A **human agent** has sensory organs such as eyes, ears, nose, tongue and skin parallel to the sensors, and other organs such as hands, legs, mouth, for effectors.
- A **robotic agent** replaces cameras and infrared range finders for the sensors, and various motors and actuators for effectors.
- A **software agent** has encoded bit strings as its programs and actions.



Problems, problem space and search

Building a system to solve a problem requires the following steps:

1. Define the problem precisely including detailed specifications and what constitutes an acceptable solution;
2. Analyze the problem thoroughly for some features may have a dominant effect on the chosen method of solution.
3. Isolate and represent the background knowledge needed in the solution of the problem.
4. Choose the best problem-solving techniques in the solution.

PROBLEM SOLVING:-

Problem solving is a process of generating solutions from observed data.

- a ‘problem’ is characterized by a set of goals,
- a set of objects, and
- a set of operations.
- A ‘**problem space**’ is an abstract space.
 - A problem space encompasses all valid states that can be generated by the application of any combination of operators on any combination of objects.
 - The problem space may contain one or more solutions. A solution is a combination of operations and objects that achieve the goals.
- A ‘**search**’ refers to the search for a solution in a problem space.
 - Search proceeds with different types of ‘search control strategies’.
 - The depth-first search and breadth-first search are the two common search strategies.

Defining the problem as a state space search

The state space representation forms the basis of most of the AI methods.

Its structure corresponds to the structure of problem solving in two important ways:

- ☐ It allows for a formal definition of a problem as the need to convert some given situation into some desired situation using a set of permissible operations.
- ☐ It permits us to define the process of solving a problem as a combination of known techniques (each represented as a rule defining a single step in the space) and search, the general technique of exploring the space to try to find some path from current state to a goal state.
- ☐ Search is a very important process in the solution of hard problems for which no more direct techniques are available.

Example: Playing Chess:- To build a program that could “play chess”, we could first have to specify the starting position of the chess board, the rules that define the legal moves, and the board positions that represent a win for one side or the other. In addition, we must make explicit the previously implicit goal of not only playing the legal game of chess but also winning the game, if possible.

Water Jug Problem:-

The state space for this problem can be described as the set of ordered pairs of integers (x,y) such that $x = 0, 1, 2, 3$ or 4 and $y = 0, 1, 2$ or 3 ; x represents the number of gallons of water in the 4-gallon jug and y represents the quantity of water in 3-gallon jug

- The start state is (0,0)
- The goal state is (2,n)

Where n may be any but it is limited to three holding from 0 to 3 gallons of water or empty.

Production rules for Water Jug Problem

- The operators to be used to solve the problem can be described as follows:

Sl No	Current state	Next State	Description
1	(x,y) if $x < 4$	(4,y)	Fill the 4 gallon jug
2	(x,y) if $y < 3$	(x,3)	Fill the 3 gallon jug
3	(x,y) if $x > 0$	(x-d, y)	Pour some water out of the 4 gallon jug
4	(x,y) if $y > 0$	(x, y-d)	Pour some water out of the 3-gallon jug
5	(x,y) if $x > 0$	(0, y)	Empty the 4 gallon jug
6	(x,y) if $y > 0$	(x,0)	Empty the 3 gallon jug on the ground
7	(x,y) if $x+y \geq 4$ and $y > 0$	(4, y-(4-x))	Pour water from the 3 – gallon jug into the 4 –gallon jug until the 4-gallon jug is full
8	(x, y) if $x+y \geq 3$ and $x > 0$	(x-(3-y), 3)	Pour water from the 4-gallon jug into the 3-gallon jug until the 3-gallon jug is full
9	(x, y) if $x+y \leq 4$ and $y > 0$	(x+y, 0)	Pour all the water from the 3-gallon jug into the 4-gallon jug
10	(x, y) if $x+y \leq 3$ and $x > 0$	(0, x+y)	Pour all the water from the 4-gallon jug into the 3-gallon jug
11	(0,2)	(2,0)	Pour the 2 gallons from 3-gallon jug into the 4-gallon jug
12	(2,y)	(0,y)	Empty the 2 gallons in the 4-gallon jug on the ground

Gallons in the 4-gallon jug	Gallons in the 3-gallon jug	Rule applied
0	0	2
0	3	9
3	0	2
3	3	7
4	2	5 or 12
0	2	9 Or 11

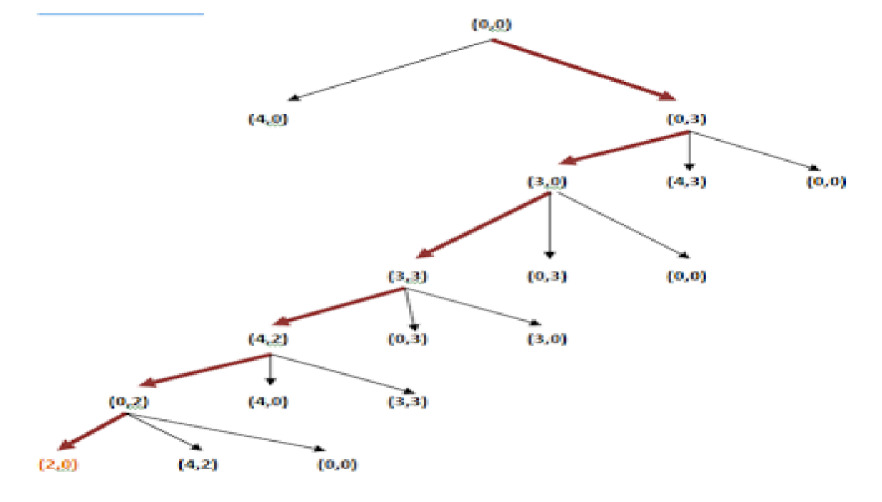
- The problem solved by using the production rules in combination with an appropriate control strategy, moving through the problem space until a path from an initial state to a goal state is found.

In this problem solving process, search is the fundamental concept. For simple problems it is easier to achieve this goal by hand but there will be cases where this is far too difficult. **Problem: You are given two jugs, a 4-gallon one and a 3-gallon one. Neither has any measuring mark on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into the 4-gallon jug.**

Solution: The state space for this problem can be described as the set of ordered pairs of integers (x,y) where,

- X represents the quantity of water in the 4-gallon jug $X = 0, 1, 2, 3, 4$
- Y represents the quantity of water in 3-gallon jug $Y = 0, 1, 2, 3$
- Start State: (0,0)
- Goal State: (2,0)
- Generate production rules for the water jug problem

State Space Tree:



Production Systems

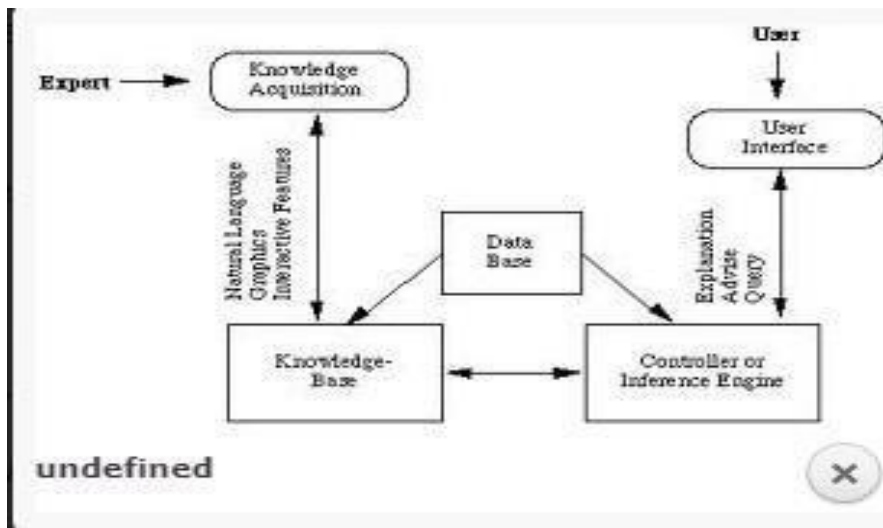
A production system is nothing but a set of rules, each rule consists of LHS and RHS where LHS patterns determines the applicability of the rules and a RHS describe the operations to be performed if the rule applied.

Production systems can be defined as a kind of cognitive architecture, in which knowledge is represented in the form of rules. So, a system that uses this form of knowledge representation is called a production system.

- Production systems are rules of the form $C \rightarrow A$ where L.H.S. is known as condition and R.H.S. is action.
- LHS describe applicability of rule and RHS describe operation to be performed.
- If one adopts a system with production rule and rule interpreter then that system is known as production system.

A production system consists of four basic components:

1. A set of rules Of the form $C_i \rightarrow A_i$ where C_i is the condition part and A_i is the action part. The condition determines when a given rule is applied, and the action determines what happens when it is applied.
2. One or more **knowledge databases** that contain whatever information is relevant for the given problem. Some parts of the database may be permanent, while others may temporary and only exist during the solution of the current problem. The information in the databases may be structured in any appropriate manner.
3. A **control strategy** that determines the order in which the rules are applied to the database, and provides a way of resolving any conflicts that can arise when several rules match at once.
4. A **rule applicer** which is the computational system that implements the control strategy and applies the rules.



Classifications of Production Systems

Production system describes the operations that can be performed in a search for a solution to the problem.

1. A monotonic production system
 2. A non-monotonic production system
 3. A partially commutative production system
 4. A commutative production system.
- **A monotonic production system:**
 - It is a production system in which the application of a rule never prevents the later application of another rule that could also have been applied at the time first rule was selected. Adding knowledge does not reduce the set of proposition that can be derived.
 - All facts are already present to solve a problem.
 - Not effective for real time system(as facts changes).
 - **A non-monotonic production system** is the one in which the application of a particular sequence of rules transforms state X into state Y, then any permutation of those rules that is allowable also transforms state x into state Y.
 - **Partially commutative production system:** A partially commutative production system is a production system with the property that if the application of a particular sequence of rules transforms state x into state y then any permutation of those rules that is allowable (i.e. each rules preconditions are satisfied when it is applied) also transforms state x into state y. Partially commutative, monotonic production systems are useful for solving ignorable problems.
 - **A commutative production system:** A commutative production system is a production system that is both monotonic and partially commutative.

Relationships:-

- Partially commutative, monotonic production systems are useful for solving ignorable problems. These systems are important for man implementation standpoint because they can be implemented without the ability to backtrack to previous states, when it is discovered that an incorrect path was followed. Such systems increase the efficiency since it is not necessary to keep track of the changes made in the search process.
- Monotonic partially commutative systems are useful for problems in which changes occur but can be reversed and in which the order of operation is not critical (ex: 8 puzzle problem).
- Production systems that are not partially commutative are useful for many problems in which irreversible changes occur, such as chemical analysis. When dealing with such systems, the order in which operations are performed is very important and hence correct decisions have to be made at the first time itself.

Advantages of production systems:-

1. Production systems provide an excellent tool for structuring AI programs.
2. Production Systems are highly modular because the individual rules can be added, removed or modified independently.
3. The production rules are expressed in a natural form, so the statements contained in the knowledge base should the recording of an expert thinking out loud.

Disadvantages of Production Systems:-

1. One important disadvantage is the facts that it may be very difficult analyze the flow of control within a production system because the individual rules don't call each other.
2. Production systems describe the operations that can be performed in a search for a solution to the problem.

Design of Search Programs

Each search process can be considered to be a tree traversal. The object of the search is to find a path from the initial state to a goal state using a tree. The number of nodes generated might be huge; and in practice many of the nodes would not be needed. The secret of a good search routine is to generate only those nodes that are likely to be useful, rather than having a precise tree. The rules are used to represent the tree implicitly and only to create nodes explicitly if they are actually to be of use.

The following issues arise when searching:

- The tree can be searched forward from the initial node to the goal state or backwards from the goal state to the initial state.
- To select applicable rules, it is critical to have an efficient procedure for matching rules against states.
- How to represent each node of the search process? This is the knowledge representation problem or the frame problem. In games, an array suffices; in other problems, more complex data structures are needed.
- Finally in terms of data structures, considering the water jug as a typical problem do we use a graph or tree? The breadth-first structure does take note of all nodes generated but the depth-first one can be modified.

Heuristic search and function:-

The purpose of heuristic function is to guide the search process in the most profitable path among all that are available.

For finding a solution, by using the heuristic technique, one should carry out the following steps:

1. **Add domain**—specific information to select what is the best path to continue searching along.
2. Define a heuristic function $h(n)$ that estimates the ‘goodness’ of a node n . Specifically, $h(n)$ = estimated cost(or distance) of minimal cost path from n to a goal state.
3. The term, heuristic means ‘serving to aid discovery’ and is an estimate, based on domain specific information that is computable from the current state description of how close we are to a goal.

Example:-

Finding a route from one city to another city is an example of a search problem in which different search orders and the use of heuristic knowledge are easily understood.

1. **State:** The current city in which the traveller is located.
2. **Operators:** Roads linking the current city to other cities.
3. **Cost Metric:** The cost of taking a given road between cities.
4. **Heuristic information:** The search could be guided by the direction of the goal city from the current city, or we could use airline distance as an estimate of the distance to the goal.

Heuristic search techniques

For complex problems, the traditional algorithms, presented above, are unable to find the solution within some practical time and space limits. Consequently, many special techniques are developed, using heuristic functions.

- Blind search is not always possible, because it requires too much time or Space (memory).
- Heuristics are rules of thumb; they do not guarantee a solution to a problem.
- Heuristic Search is a weak technique but can be effective if applied correctly; it requires domain specific information.

Characteristics of heuristic search

- Heuristics are knowledge about domain, which help search and reasoning in its domain.

- Heuristic search incorporates domain knowledge to improve efficiency over blind search.

- Heuristic is a function that, when applied to a state, returns value as estimated merit of state, with respect to goal.
 - Heuristics might (for reasons) underestimate or overestimate the merit of a state with respect to goal.
 - Heuristics that underestimate are desirable and called admissible.
- Heuristic evaluation function estimates likelihood of given state leading to goal state.
- Heuristic search function estimates cost from current state to goal, presuming function is efficient.

Best First Search

- Best first search is an instance of graph search algorithm in which a node is selected for expansion based on evaluation function $f(n)$. Traditionally, the node which is the lowest evaluation is selected for the explanation because the evaluation measures distance to the goal.
- Best first search can be implemented within general search frame work via a priority queue, a data structure that will maintain the fringe in ascending order of f values.
- This search algorithm serves as combination of depth first and breadth first search algorithm. Best first search algorithm is often referred greedy algorithm this is because they quickly attack the most desirable path as soon as its heuristic weight becomes the most desirable.

Concept:

Step 1: Traverse the root node

Step 2: Traverse any neighbour of the root node, that is maintaining a least distance from the root node and insert them in ascending order into the queue.

Step 3: Traverse any neighbour of neighbour of the root node, that is maintaining a least distance from the root node and insert them in ascending order into the queue

Step 4: This process will continue until we are getting the goal node

Algorithm:

Step 1: Place the starting node or root node into the queue.

Step 2: If the queue is empty, then stop and return failure.

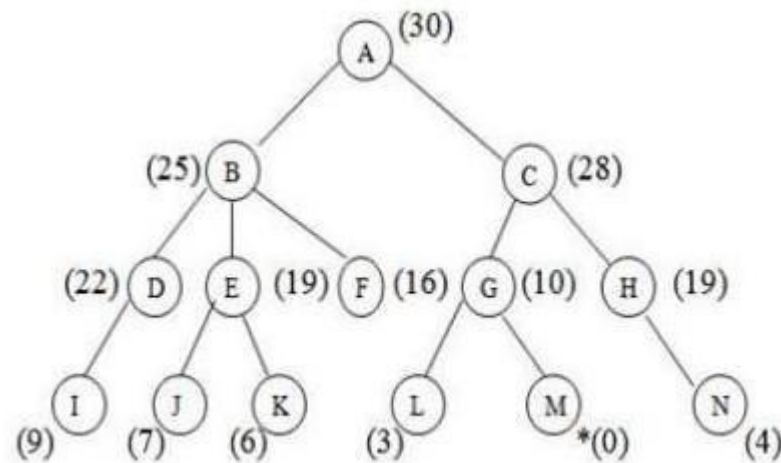
Step 3: If the first element of the queue is our goal node, then stop and return success.

Step 4: Else, remove the first element from the queue. Expand it and compute the estimated goal distance for each child. Place the children in the queue in ascending order to the goal distance.

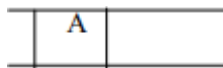
Step 5: Go to step-3

Step 6: Exit.

Implementation:



Step 1: Consider the node A as our root node. So the first element of the queue is A which is not our goal node, so remove it from the queue and find its neighbour that are to inserted in ascending order.



Step 2: The neighbours of A are B and C. They will be inserted into the queue in ascending order.



Step 3: Now B is on the FRONT end of the queue. So calculate the neighbours of B that are maintaining a least distance from the roof.



Step 4: Now the node F is on the FRONT end of the queue. But as it has no further children, so remove it from the queue and proceed further.



Step 5: Now E is the FRONT end. So the children of E are J and K. Insert them into the queue in ascending order.



Step 6: Now K is on the FRONT end and as it has no further children, so remove it and proceed further



Step7: Also, J has no corresponding children. So remove it and proceed further.



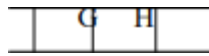
Step 8: Now D is on the FRONT end and calculates the children of D and put it into the queue.



Step9: Now I is the FRONT node and it has no children. So proceed further after removing this node from the queue.

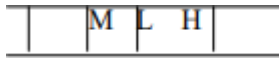


Step 10: Now C is the FRONT node .So calculate the neighbours of C that are to be inserted in ascending order into the queue.



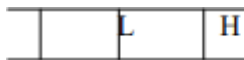
C

Step 11: Now remove G from the queue and calculate its neighbour that is to insert in ascending order into the queue.



G

Step 12: Now M is the FRONT node of the queue which is our goal node. So stop here and exit.



M

Advantage:

- It is more efficient than that of BFS and DFS.
- Time complexity of Best first search is much less than Breadth first search.
- The Best first search allows us to switch between paths by gaining the benefits of both breadth first and depth first search.
- Because, depth first is good because a solution can be found without computing all nodes and Breadth first search is good because it does not get trapped in dead ends.

Disadvantages: Sometimes, it covers more distance than our consideration.

Generate and Test Strategy

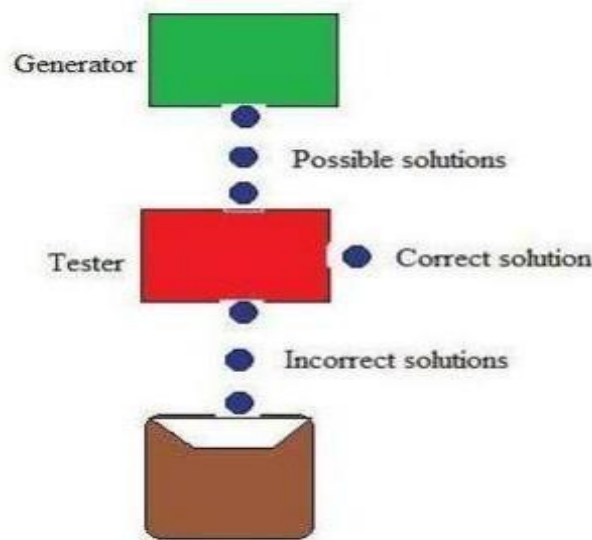
Generate-And-Test Algorithm

Generate-and-test search algorithm is a very simple algorithm that guarantees to find a solution if done systematically and there exists a solution.

Algorithm: Generate-And-Test

1. Generate a possible solution.
2. Test to see if this is the expected solution.
3. If the solution has been found quit else go to step 1.

Potential solutions that need to be generated vary depending on the kinds of problems. For some problems the possible solutions may be particular points in the problem space and for some problems, paths from the start state.



Generate-and-test, like depth-first search, requires that complete solutions be generated for testing. In its most systematic form, it is only an exhaustive search of the problem space. Solutions can also be generated randomly but solution is not guaranteed.

Systematic Generate-And-Test

- While generating complete solutions and generating random solutions are the two extremes there exists another approach that lies in between.
- The approach is that the search process proceeds systematically but some paths that unlikely to lead the solution are not considered. This evaluation is performed by a heuristic function.
- Depth-first search tree with backtracking can be used to implement systematic generate-and-test procedure. As per this procedure, if some intermediate states are likely to appear often in the tree, it would be better to modify that procedure to traverse a graph rather than a tree.

Generate-And-Test And Planning

Exhaustive generate-and-test is very useful for simple problems. But for complex problems even heuristic generate-and-test is not very effective technique. But this

may be made effective by combining with other techniques in such a way that the space in which to search is restricted.

An AI program DENDRAL, for **example**, uses plan-Generate-and-test technique. First, the planning process uses constraint-satisfaction techniques and creates lists of recommended and contraindicated substructures.

Then the generate-and-test procedure uses the lists generated and required to explore only a limited set of structures. Constrained in this way, generate-and-test proved highly effective.

A major weakness of planning is that it often produces inaccurate solutions as there is no feedback from the world. But if it is used to produce only pieces of solutions then lack of detailed accuracy becomes unimportant.

Constraint Satisfaction Problems:-

It is defined by a set of variables $x_1, x_2, \dots, x_n$ and set of conditions $c_1, c_2, \dots, c_n$ such that each of the variable x_i has a non-empty domain D_i of possible values.

CSP is defined by 3 components (X, D, C) :

- State: a set of variables X , each X_i , with values from domain D_i .
- Goal test: a set of constraints C , each C_i involves some subset of the variables and specifies the allowable combinations of values for that subset {
- Each constraint C_i consists of a pair $\langle \text{scope}, \text{rel} \rangle$, where scope is a tuple of variables and rel is the relation, either represented explicitly or abstractly
- X_1 and X_2 both have the domain $\{A, B\}$

- **Constraints:**

$\langle (X_1, X_2), [(A, B), (B, A)] \rangle$, OR
 $\langle (X_1, X_2), X_1 \neq X_2 \rangle$

1. Cryptarithmic problem
2. N-queen problem
3. Cross word problem
4. Map coloring problem

After WA- R	R	G, B	G, B	R,G,B	R,G,B	R,G,B	R,G,B
After NT-G	R	G	B	R	G	R	R,G,B
After T- R	R	G	B	R	G	R	R

Variables: $X = \{WA, NT, Q, NSW, V, SA, T\}$

Domains: $D_i = \{\text{red, green, blue}\}$

Constraints: adjacent regions must have different colors

Solution:- $\{WA = \text{red}, NT = \text{green}, Q = \text{red}, NSW = \text{green}, V = \text{red}, SA = \text{blue}, T = \text{red}\}$.

Criptarithmic problem:-

- Cryptarithmic Problem “It is an arithmetic problem which is represented in letters. It involves the decoding of digit represented by a character.
- It is in the form of some arithmetic equation where digits are distinctly represented by some characters.
- The problem requires finding of the digit represented by each character. Assign a decimal digit to each of the letters in such a way that the answer to the problem is correct.
- If the same letter occurs more than once, it must be assigned the same digit each time. No two different letters may be assigned the same digit”

Instructions for Cryptarithmic Problems

1. Each variable should have unique and distinct value.
2. Alphabets can only take up values between 0-9.
3. Each letter, symbol represents only one digit throughout the problem.
4. You have to find the value of each letter in the Cryptarithmic.
5. Decoded numbers can't begin with 0, for example, 0813.
6. 19 is the max value with a carryover for two one-digit number in the same column.
7. Carry over can only be 1.

8. After replacing letter by their digits, the resulting arithmetic operations must be correct.

1) SEND + MORE = MONEY

$$\begin{array}{r}
 \begin{array}{cccccc}
 & 5 & 4 & 3 & 2 & 1 \\
 & \text{S} & \text{E} & \text{N} & \text{D} & \\
 + & \text{M} & \text{O} & \text{R} & \text{E} & \\
 & & \text{c}_3 & \text{c}_2 & \text{c}_1 & \\
 \hline
 \end{array}
 \end{array}$$

M O N E Y

1. From Column 5, $M=1$, since it is only carry-over possible from sum of 2 single digit number in column 4.
2. To produce a carry from column 4 to column 5 'S + M' is atleast 9 so ' $S=8 \text{ or } 9$ ' so ' $S+M=9 \text{ or } 10$ ' & so ' $O = 0 \text{ or } 1$ '. But ' $M=1$ ', so ' $O = 0$ '.
3. If there is carry from Column 3 to 4 then ' $E=9$ ' & so ' $N=0$ '. But ' $O = 0$ ' so there is no carry & ' $S=9$ ' & ' $c_3=0$ '.
4. If there is no carry from column 2 to 3 then ' $E=N$ ' which is impossible, therefore there is carry & ' $N=E+1$ ' & ' $c_2=1$ '.
5. If there is carry from column 1 to 2 then ' $N+R=E \bmod 10$ ' & ' $N=E+1$ ' so ' $E+1+R=E \bmod 10$ ', so ' $R=9$ ' but ' $S=9$ ', so there must be carry from column 1 to 2. Therefore ' $c_1=1$ ' & ' $R=8$ '.
6. To produce carry ' $c_1=1$ ' from column 1 to 2, we must have ' $D+E=10+Y$ ' as Y cannot be 0/1 so D+E is atleast 12. As D is atleast 7 & E is atleast 5 (D cannot be 8 or 9 as it is already assigned). N is atleast 7 & ' $N=E+1$ ' so ' $E=5 \text{ or } 6$ '.
7. If E were 6 & D+E atleast 12 then D would be 7, but ' $N=E+1$ ' & N would also be 7 which is impossible. Therefore ' $E=5$ ' & ' $N=6$ '.
8. D+E is atleast 12 for that we get ' $D=7$ ' & ' $Y=2$ '.

SOLUTION:

$$\begin{array}{r}
 \begin{array}{cccccc}
 & 9 & 5 & 6 & 7 & \\
 + & 1 & 0 & 8 & 5 & \\
 \hline
 1 & 0 & 6 & 5 & 2 &
 \end{array}
 \end{array}$$

VALUES:

S=9
 E=5
 N=6
 D=7
 M=1
 O=0
 R=8
 Y=2

Hill Climbing:-

Hill climbing is an example of an informed search method because it uses information about the search space to search in a reasonably efficient manner. The Hill climbing search always moves towards the goal. Using heuristics it finds which direction will take it closest to the goal. The name hill climbing is derived from simulating the situation of a person climbing the hill. A hiker is lost halfway up/down a mountain at night. His camp is at the top of the mountain. Even though it is dark, the hiker knows that every step he takes up the mountain is a step towards his goal.

The algorithm for Hill climbing is as follows:

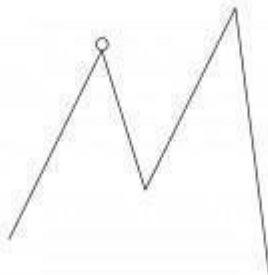
1. Evaluate the initial state, if it is goal state quit otherwise make current state as initial state.
2. Select a new operator that could be applied to this state and generate a new state.
3. Evaluate the new state
 - If this new state is closer to the goal state, than current state make the new state as the current state.
 - If it is not better ignore this state and proceed with current state.
 - If the current state is goal state or no new operators are available, quit. Otherwise repeat from 2.

Problems with Hill climbing

There are three problems encountered in hill climbing technique:

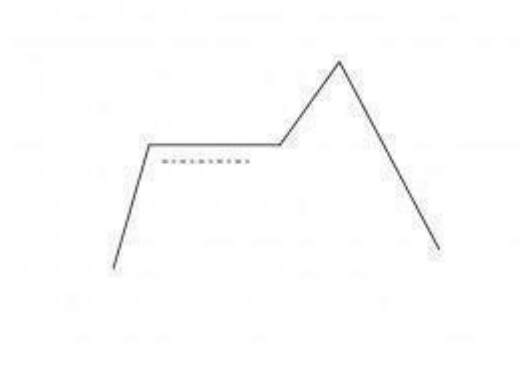
1. Local maxima

It is a state which is better than all of its neighbors but is not better than some other states which are farther away. It is a state where we have climbed to the top of the hill, and missed the mountain.



Solution to the problem

- i. One possible solution is backtracking.
We can backtrack to some earlier node and try to go in a different direction to attain the global peak. We can maintain a list of paths almost taken and go back to one of them if the path that was taken leads to a dead end.
 - ii. Another solution can be a list of promising plan.
- 2. Plateau-**It is a state where everything around is about as good as where we are currently. In other words a flat area of the search space in which all neighboring states have the same value.



Solution to the problem

- iii. A big jump in some direction can be done in order to get to a new section of search space. This method is recommended as in a plateau all neighboring points have the same value.
- iv. Another solution is to apply small steps several times in the same direction. This depends on the rules available.

3. Ridges

It is a special kind of local maximum. It is an area of the search space which is higher than the surrounding areas and that itself has a slope. We cannot travel the ridge by single moves as the orientation of the high region compared to the set of available moves makes it impossible.



Solution to the problem

- v. Trying different paths at the same time is a solution. We can apply two or more rules before doing the test. This corresponds to moving in several directions at once.
- vi. Bidirectional search can be useful here.

Advantages of Hill Climbing

- It can be used in continuous as well as discrete domains

Disadvantages of Hill Climbing

- It is not an efficient method.
- It is not suited to problems where the value of the heuristic function drops off suddenly when the solution may be in sight.
- It is a local method as it looks at the immediate solution and decides about the next step to be taken rather than exploring all consequences before taking a move.

Problem Reduction:-

A* algorithm:-

It is best known form of Best First Search. It avoids path that are already expensive but expands most promising path first.

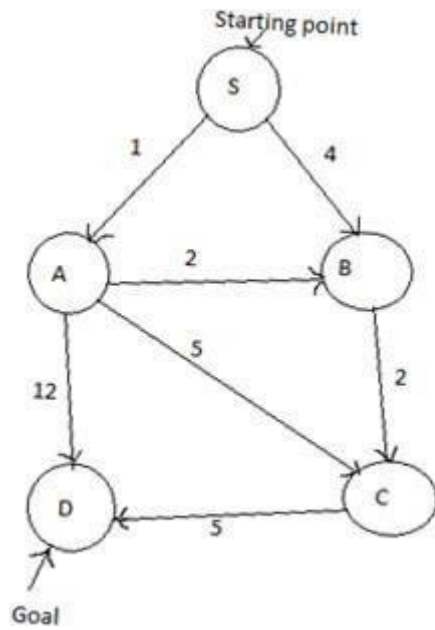
$$f(n) = g(n) + h(n)$$

$g(n)$ □ the cost to reach the

node $h(n)$ □ heuristic value

$f(n)$ □ estimated total path to reach goal.

Example:-



Heuristic Values	
S	7
A	6
B	2
C	1
D	0

$$S \rightarrow [f(n)=g(n)+h(n)]$$

$$f(n)=0+7$$

$$=7$$

$$S \rightarrow A$$

$$g(n) = 1, h(n) = 6$$

$$f$$

$$($$

$$n$$

$$)$$

$$=$$

$$g$$

$$($$

$$n$$

$$)$$

$$+$$

$$S \rightarrow B$$

h

(

n

)

f

(

n

)

=

1

+

6

f(n) = 7

h

(

n

)

f

(

n

)

=

4

+

2

f(n) = 6

g(n) = 4, h(n) = 2

f

(

n

)

=

g

(

n

)

+

$$S \sqcup B \sqcup C$$

$$g(n) = 6, h(n) = 1$$

f

(

n

)

=

g

(

n

)

+

h

(

n

)

f

(

n

)

=

6

+

1

f

$$\begin{aligned} & (\\ & \mathbf{n} \\ &) \\ & = \\ & 7 \\ & \mathbf{S} \\ & \square \\ & \mathbf{A} \\ & \square \\ & \mathbf{B} \\ & \mathbf{g} \\ & (\\ & \mathbf{n} \\ &) \\ & = \\ & 3 \\ & , \\ & \mathbf{h} \\ & (\\ & \mathbf{n} \\ &) \\ & = \\ & 2 \end{aligned}$$

$$\begin{aligned} & \mathbf{f} \\ & (\\ & \mathbf{n} \\ &) \\ & = \\ & \mathbf{g} \\ & (\\ & \mathbf{n} \\ &) \\ & + \\ & \mathbf{h} \\ & (\\ & \mathbf{n} \\ &) \\ & \mathbf{f} \\ & (\\ & \mathbf{n} \\ &) \\ & = \\ & 3 \\ & + \\ & 2 \\ & \mathbf{f} \\ & (\\ & \mathbf{n} \end{aligned}$$

$$) = 5 S \left[A \left[C g (n) \right] = 6 , h (n) \right] = 1 f ($$

$$n) = g (n) + h (n) f (n) =$$

7
S
□
A
□
D
g
(
n
)
=
1
3
,
h
(
n
)
=
0
f
(

n
)
=
g
(
n
)
+
h
(
n
)
f
(
n
)
=
1
3
+
0
f(n) = 13

$$(S \sqcup A \sqcup B) \sqcup C$$

$$g(n) = 5, h(n) = 1$$

$$f(n)=g(n)+h(n)$$

$$f(n) = 5 + 1$$

$$f(n) = 6$$

$$(S \cup A \cup B \cup$$

$$C) \cup D$$

$$g(n) = 8, h(n) = 0$$

$$f(n) = g(n) + h(n)$$

$$) f(n) = 8 + 0$$

$$f(n) =$$

$$8(S \cup B \cup C) \cup D$$

$$g(n) = 9, h(n) = 0$$

$$f(n) = g(n) + h(n)$$

$$) f(n) = 9 + 0$$

$$f(n) =$$

$$9(S \cup A \cup C) \cup D$$

$$g(n) = 9, h(n) = 0$$

$$f(n) = g(n) + h(n)$$

$$) f(n) = 9 + 0$$

$$f(n) = 9$$

$S \cup A$	7
$S \cup B$	6

$S \cup B \cup C$	7
$S \cup A \cup C$	7
$S \cup A \cup B$	5
$S \cup A \cup D$	13
$(S \cup A \cup B) \cup C$	6
$(S \cup A \cup B \cup C) \cup D$	8
$(S \cup B \cup C) \cup D$	9
$(S \cup A \cup C) \cup D$	9

